



THE OHIO STATE UNIVERSITY

Progress report: Estimating External Travel Using Purchased Third-Party Data

State Job # 134877
Statewide Planning and Research
Ohio Department of Transportation

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Problem statement

External cordon travel surveys

Personal and commercial travel entering, leaving, and passing through a study area

Traditional methods

Intercept

Rich data, but requires extensive field staffing; disruptive

Automated vehicle counts

e.g., license plate recognition

Lack important trip characteristics



Problem statement

Archived travel data (ATD)

Mobile communications, GPS

Archived by private companies

Airsage

Streetlight

American Transportation Research Institute
(ATRI)

A low cost and effective replacement or
complement to traditional cordon survey
methods?





Objectives

Research questions

1. Does ATD offer improvements or limitations that enhance or limit traditional travel demand forecasting performance?
2. What ATD data specifications should be applied to maximize value?
3. How can ATD be applied in traditional travel demand forecasting models to make maximum its strengths and minimize its limitations?

At the completion of this project, ODOT will have the necessary information to move forward with ATD, continue with traditional methods or implement an integrated approach



Work plan

Task 1: Research Mobilization and Data Acquisition

- Review methods and applications of ATD in transportation planning
- Assess ODOT's modeling standards and cordon survey requirements
- Acquisition of ATD (AirSage, Streetlight, ATRI)

Task 2: ATD assessment

- Study area: External cordon study in Lima (2008, 2009, 2011)
- Trip length; Trip purpose; EE flows; IE/EI flows; Traffic volume at external stations

Task 3: Recommendations for Future Ohio Model Areas

- Recommendations on future ODOT cordon travel surveys



Research context

Origin-Destination estimation from ATD

Step 1: Extract trajectories from device data

Network-based: Probe data, call data records (CDRs) with timestamps and service area IDs

GPS: Timestamps and locations at regular intervals

Step 2: Scale sample to population

Device-per vehicle equivalent (DVE): Number of occupants in the vehicle, market share of the network operator, and the likelihood that a device is switched on

Vehicle-per-device equivalent (VDE): Conditional probability of cell phone ownership using income and age distribution obtained from census



Research context

ATD advantages

- ATD sample size is typically larger
- ATD can be collected over a broader range of locations and times
- ATD collection cost is lower because it makes use of pre-installed infrastructure
- ATD can be updated in shorter time intervals

ATD disadvantages

- Not based on a random sampling frame
- Not designed for travel modeling purposes
- Intrinsic inaccuracy of location information (service network)

ADT has advantages, but requires quality assessment due to inherent biases



Research context

ADT quality measures

1. Benchmark OD table available

- Mean absolute error (MAE)
- Root mean square error (RMSE)
- Correlation analysis
- Visualization/mapping

2. Benchmark OD table unavailable

- (Other estimates available, e.g., link counts)
- Maximum possible relative error (MPRE)
- Expected relative error (ERE)
- Total demand scale (TDS)

We assumed Case 1:

- We had an OD table for Allen Co.
- Case 2 requires complex modeling of error in the benchmark table



Research context

Similar research efforts

- NCHRP 08-95 project: Use of Cellular Data to Estimate Travel
- FHWA Travel Model Improvement Program (TMIP) webinar (Spring 2016)

Other application projects

- **2012**: South Alabama Regional Planning Commission (SARPC) – Mobile, AL (AirSage)
- **2012-2014**: Kentucky Transportation Cabinet – Kentucky (AirSage)
- **2013**: North Carolina Department of Transportation (NCDOT) – Asheville, NC (AirSage)
- **2014**: Napa County Transportation Planning Agency – Napa Valley, CA (Streetlight)
- **2015**: Florida Department of Transportation – Northwest Florida (AirSage)
- **2016**: West Contra Costa County – San Francisco (AirSage)



ATD assessment: Overview

Study area

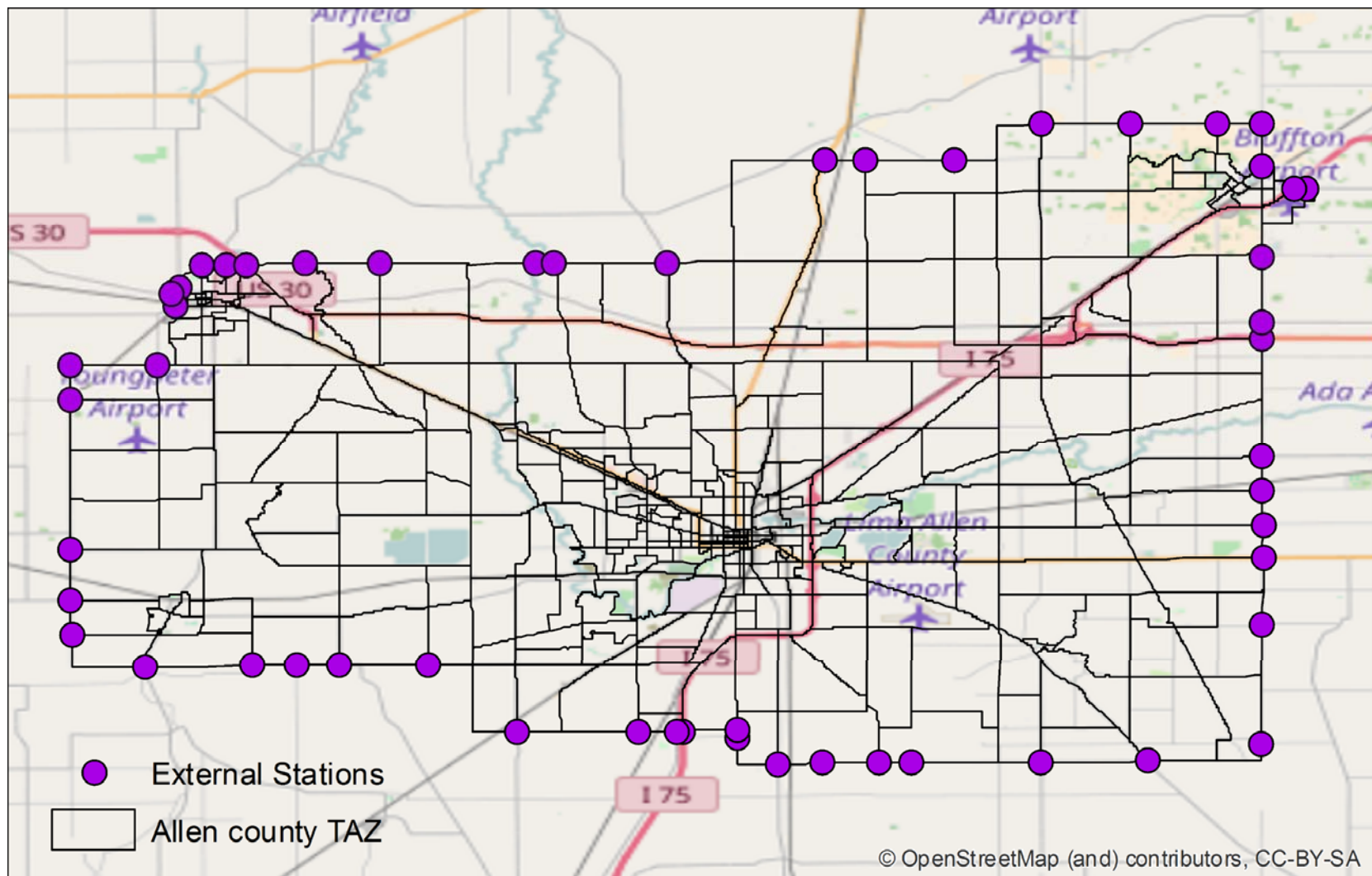
Allen County (Lima), Ohio

ODOT cordon survey (2011)

Special additional analysis
steps needed for I-75

Not possible to place cordon
on interstate







ATD assessment: Overview

Network data

ODOT layer imported to TransCad from a shape file.

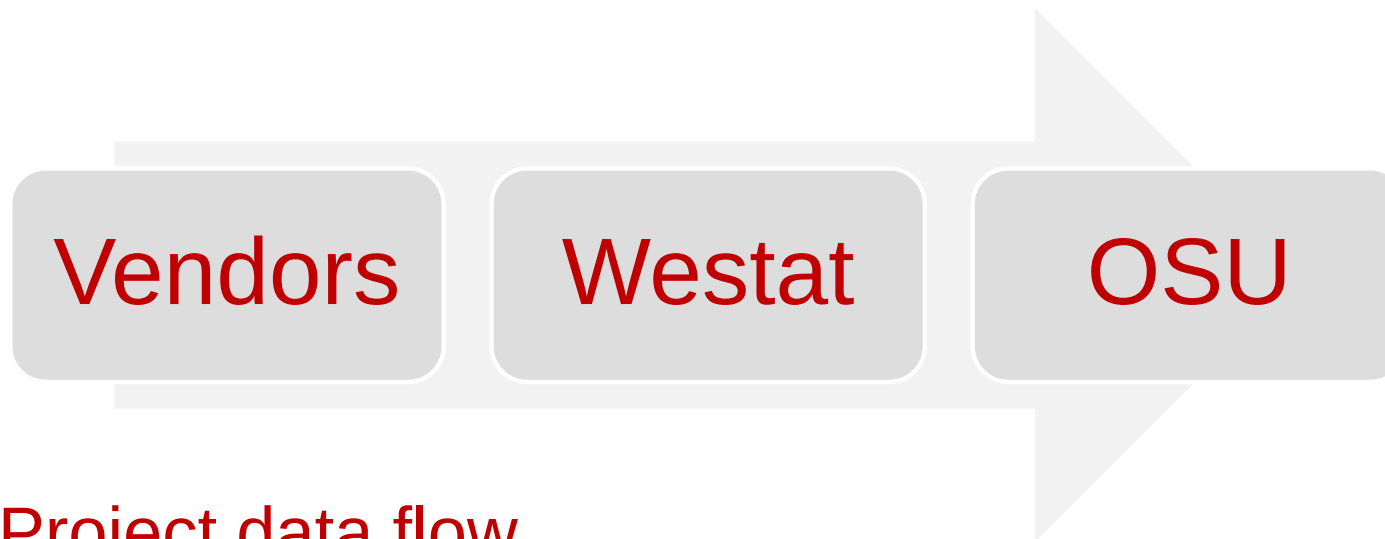
- Original uses the projection (NAD-83, Ohio South, Feet)
- Network is built from this layer and includes all the associated data from ODOT

Speed limit data

- A few records with missing speed (they are all ramps)
- We replaced these with a nominal speed of 25 mph.
- These are very short segments.



ATD assessment: Data



Project data flow

- Westat processed vendor ADT into OD tables
- OSU only received OD tables
- Process maintains confidentiality



ATD assessment: Data

Data sources:

ATRI – Heavy-duty commercial vehicle GPS data

INRIX/Streetlight – Navigation/traffic applications (pulls GPS from users), fleet / commercial vehicle GPS probe data

AirSage – Entirely from cell phone signal - triangulation from towers during phone activity

Note: all raw data is suppressed in final delivered products so that no private details are revealed for individuals or vehicles



ATD assessment: Data

Estimating external station and internal TAZ trip ends

ATRI – Provided at block group level outside of the study area. Shortest network path to estimate ingress/egress point (external station)

INRIX/Streetlight – Original data provided at the external station / TAZ level

AirSage – Uses “catchment” areas outside of study area to identify trips entering/exiting study area:

Method 1: Assign catchment trips to valid external stations and weight by station ADT

Method 2: Assign catchment trips to external stations using shortest path (between catchment area centroids and internal TAZs)

Overview of study datasets

Table 1 Data overview

ID	Data description	Trip type			Vehicle type	Trip purpose
		II	EI/IE	EE		
OD00	Unexpanded 2008 Roadside survey result	-	Y	Y	Commercial/ Personal	Work/Non-work/Truck
OD01	Expanded 2008 Roadside survey result	-	Y	Y	-	Work/Non-work/Truck
OD02	2008 Roadside survey and 2009 ALRP on IR 75	-	-	Y	-	-
OD03	2006 model external trip table	-	Y	Y	-	Work/Non-work/Truck
OD04	2011 model external trip table	-	Y	Y	-	Work/Non-work/Truck
AS01	<u>AirSage</u> (ADT weighted)	Y	Y	Y	Unknown	HH, HO, HW, OH, OO, OW, WH, WO, WW
AS02	<u>AirSage</u> (Shortest path modeled)	Y	Y	Y	Unknown	HH, HO, HW, OH, OO, OW, WH, WO, WW
ATTRUCK00	ATRI (Shortest path modeled)	Y	Y	Y	Truck	-
SC00	Streetlight Commercial (ADT weighted)	Y	Y	Y	Commercial	Unknown
SP00	Streetlight Personal (ADT weighted)	Y	Y	Y	Personal	Unknown



ATD assessment: Data

Table 3 EE trip frequency

		Trip frequency
Ground-truth	ODOT 2011 (Total)	30935
	Non-work	16164
	Work	2546
	Truck	12223
ATD	AirSage ADT weighted	15479
	AirSage Shortest path modeled	2131
	ATRI	59294
	Streetlight Commercial	1296875
	Streetlight Personal	163617

There are large scale differences between the ODOT data and the vendor ATD



ATD assessment: Results

Analysis by travel demand facet

1. Trip Length
2. Trip Purpose
3. EE Analysis
4. IE/EI Analysis
5. Traffic volume at external stations



1. Trip length

Travel time estimates

Travel times computed from the length and posted speed limit using the ODOT network data.

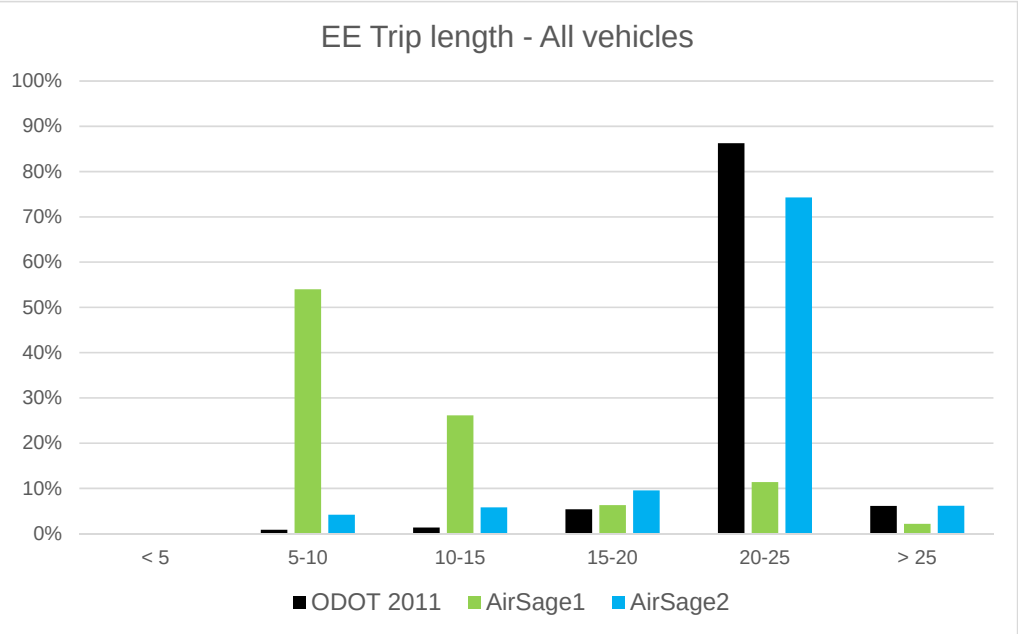
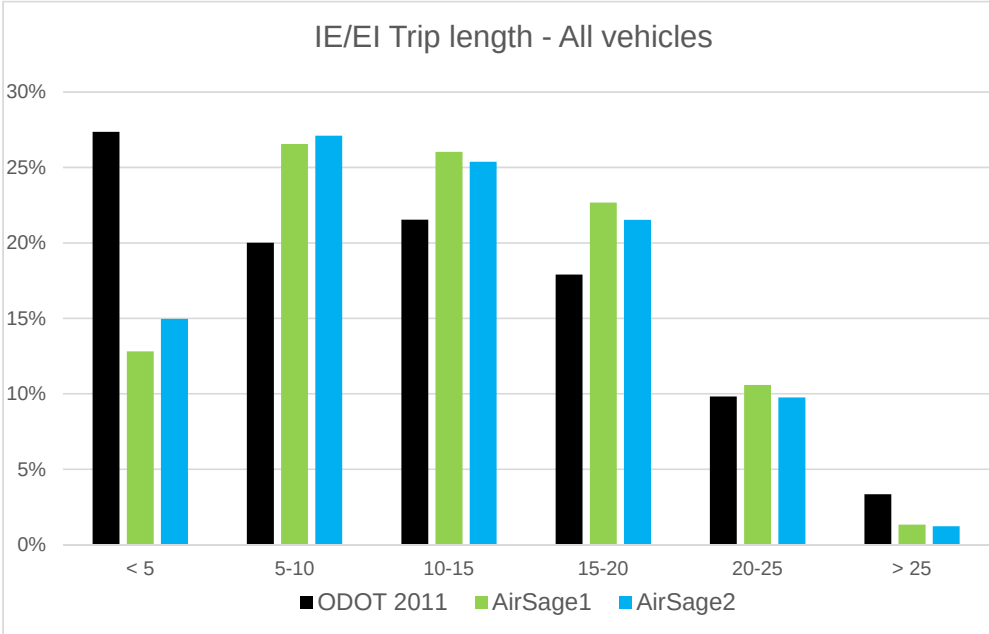
$TIME = DIST * (60 / SPD)$ where DIST is in miles, and SPD is posted speed limit in miles per hour

Trip length analysis

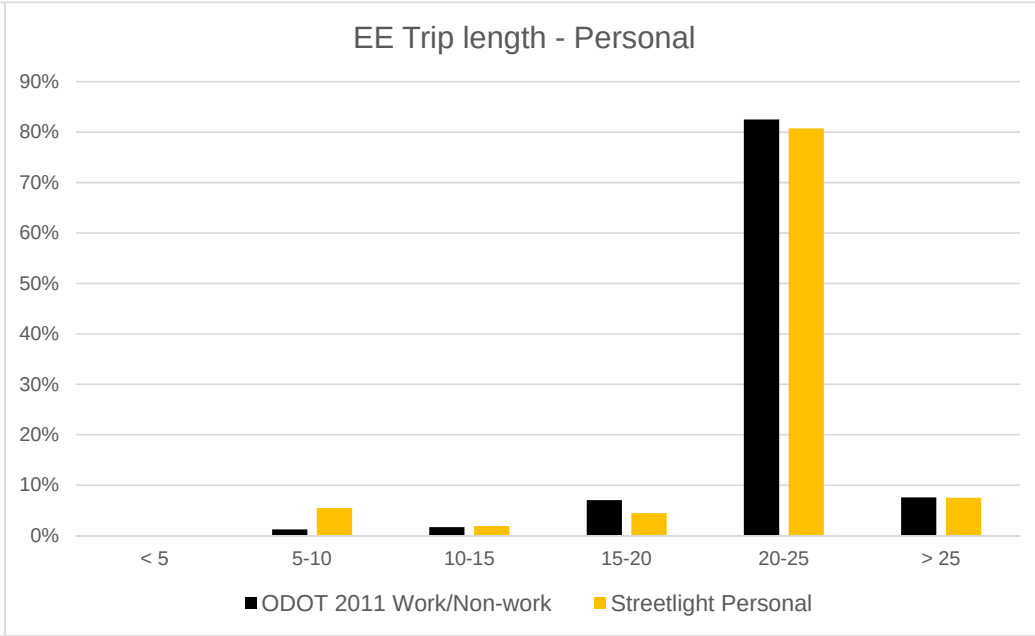
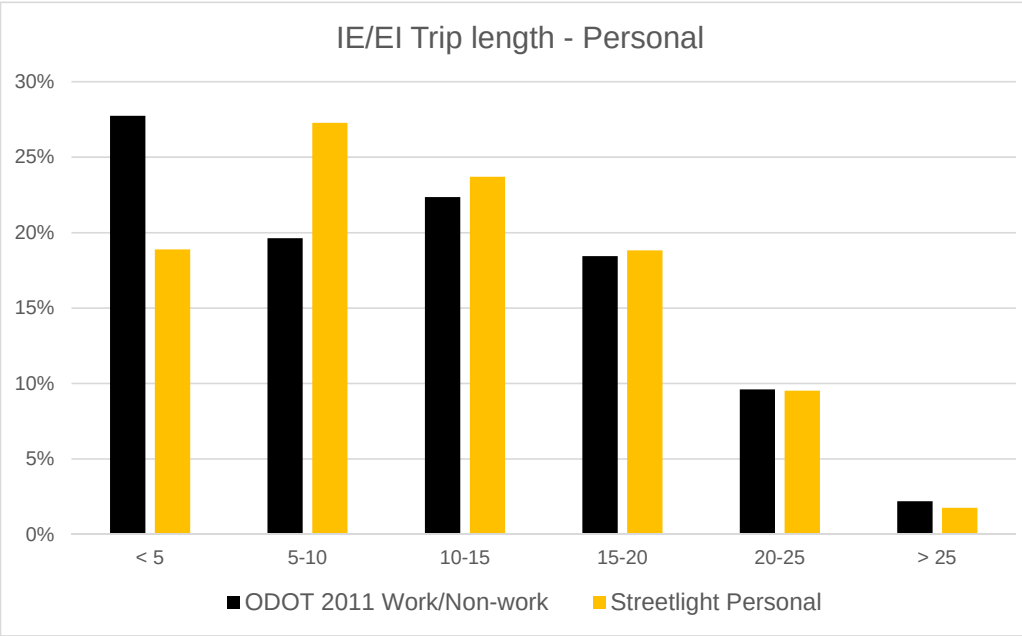
Data analyzed in a MATLAB program to categorize trips into <5, 5-10, 10-15, ... minute intervals

Selected comparisons graphed (next slides)

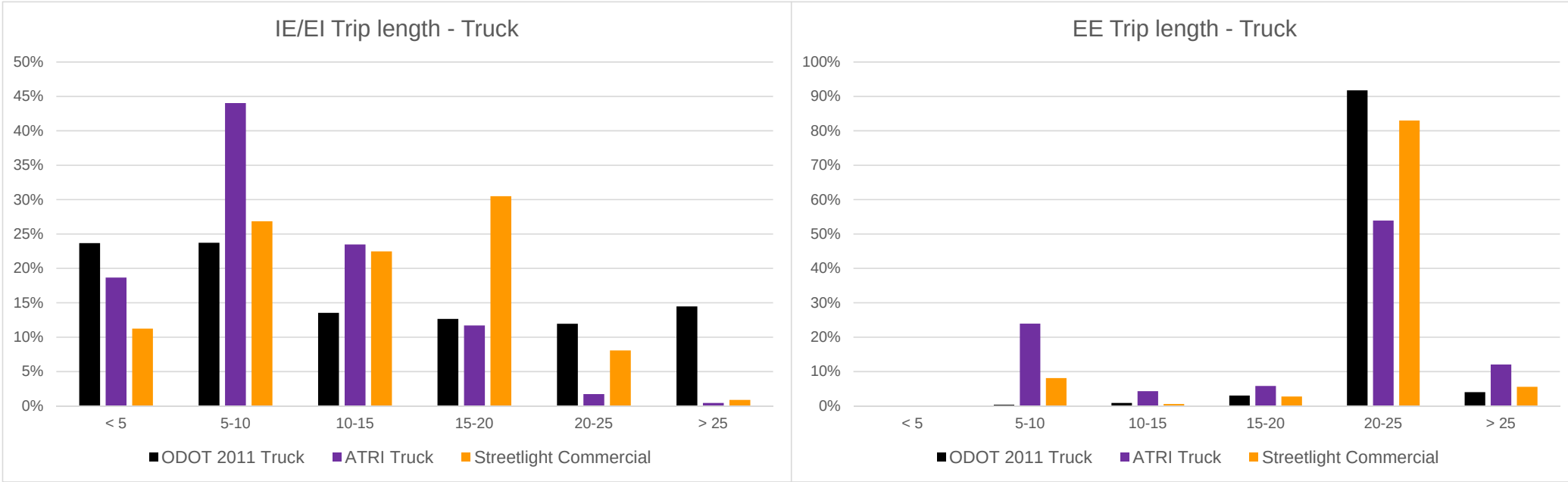
Personal vehicles: AirSage



Personal vehicles: Streetlight



Commercial vehicles: ATRI, Streetlight





2. Trip purpose

In progress...





3. EE flows

A reminder...

- Each data series is at a different scale
- This affects the error analysis
- (Next slide)

Table 3 EE trip frequency

		Trip frequency
Ground-truth	ODOT 2011 (Total)	30935
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Recall from a previous slide...



Table 2 Overall goodness of fit measures (AME, RSME and R square)

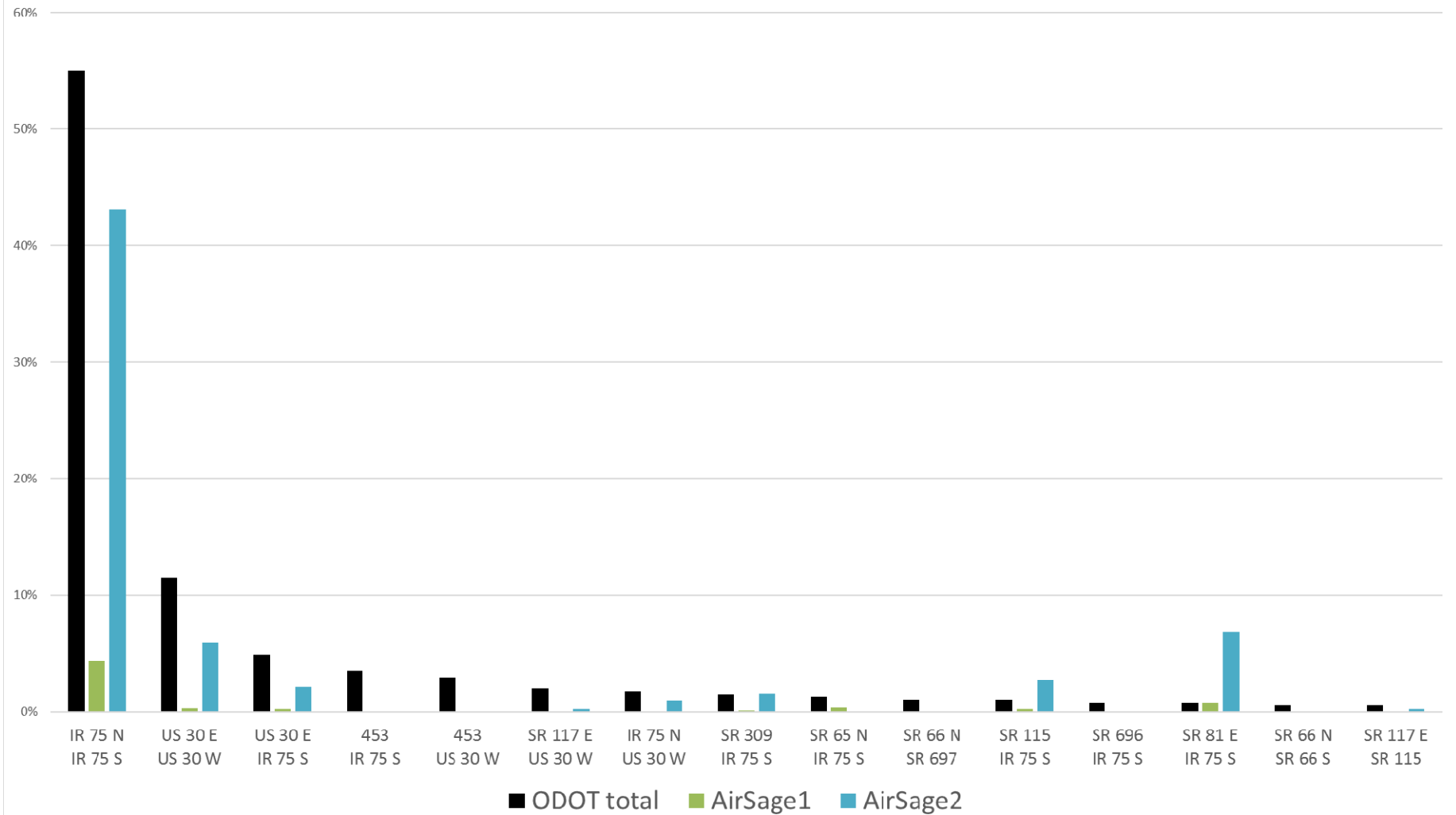
3. EE flows

Ground-truth	ATD	Absolute mean error	Root mean square error	R Square
ODOT 2011 (total)	AirSage ADT weighted	35.2	487.7	0.05
ODOT 2011 (total)	AirSage Shortest path modeled	24.2	474.0	0.93
ODOT 2011 Truck	ATRI	40.9	464.4	0.67
ODOT 2011 Truck	Streetlight Commercial	1044.2	18624.2	0.92
ODOT 2011 Non-work + work	Streetlight Personal	122.1	2269.7	0.97

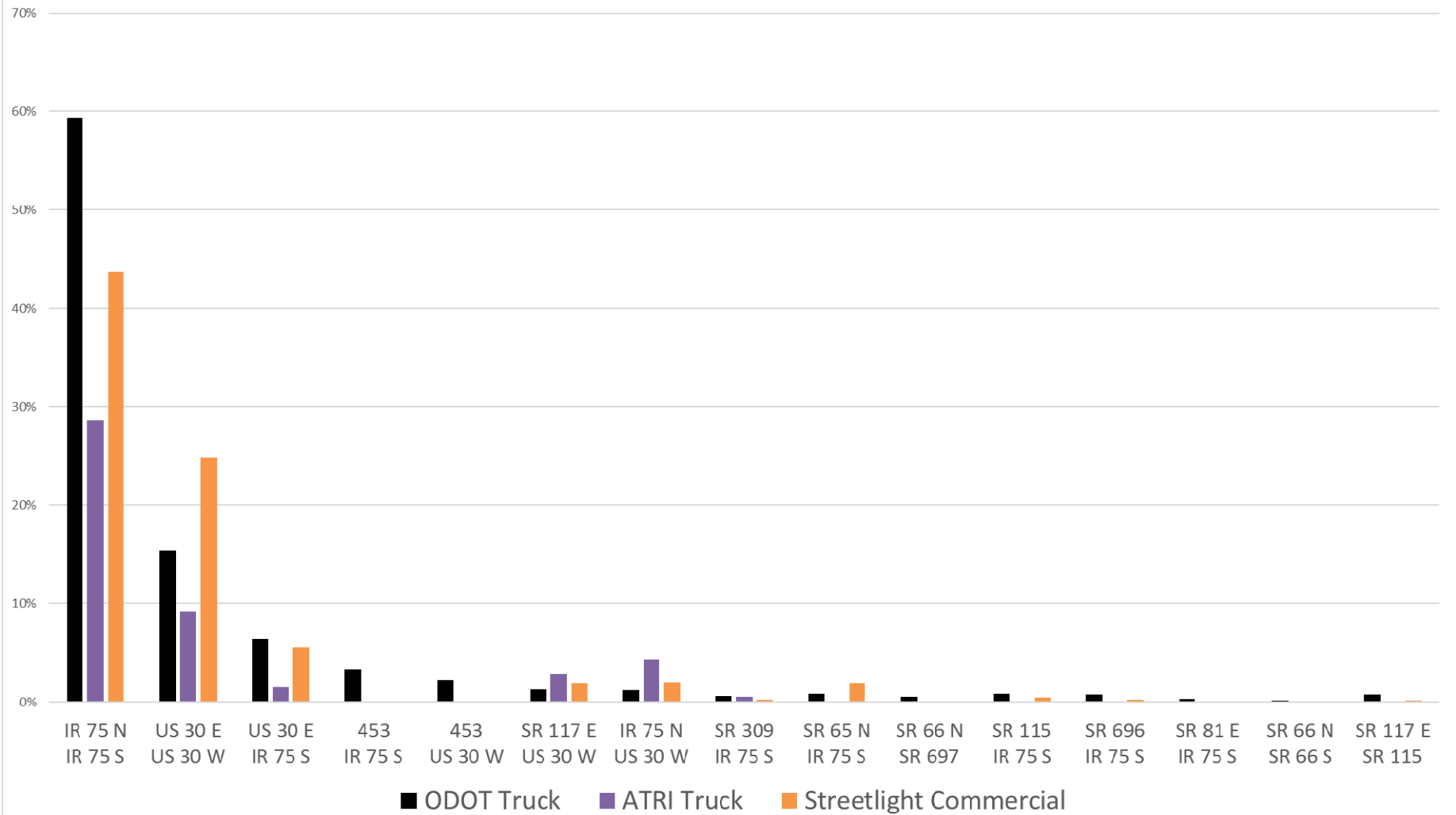
Absolute error measures suggest poor fit

Relative measures suggest better fit, implying that the relative pattern of EE flows from ATDs are similar to ODOT data

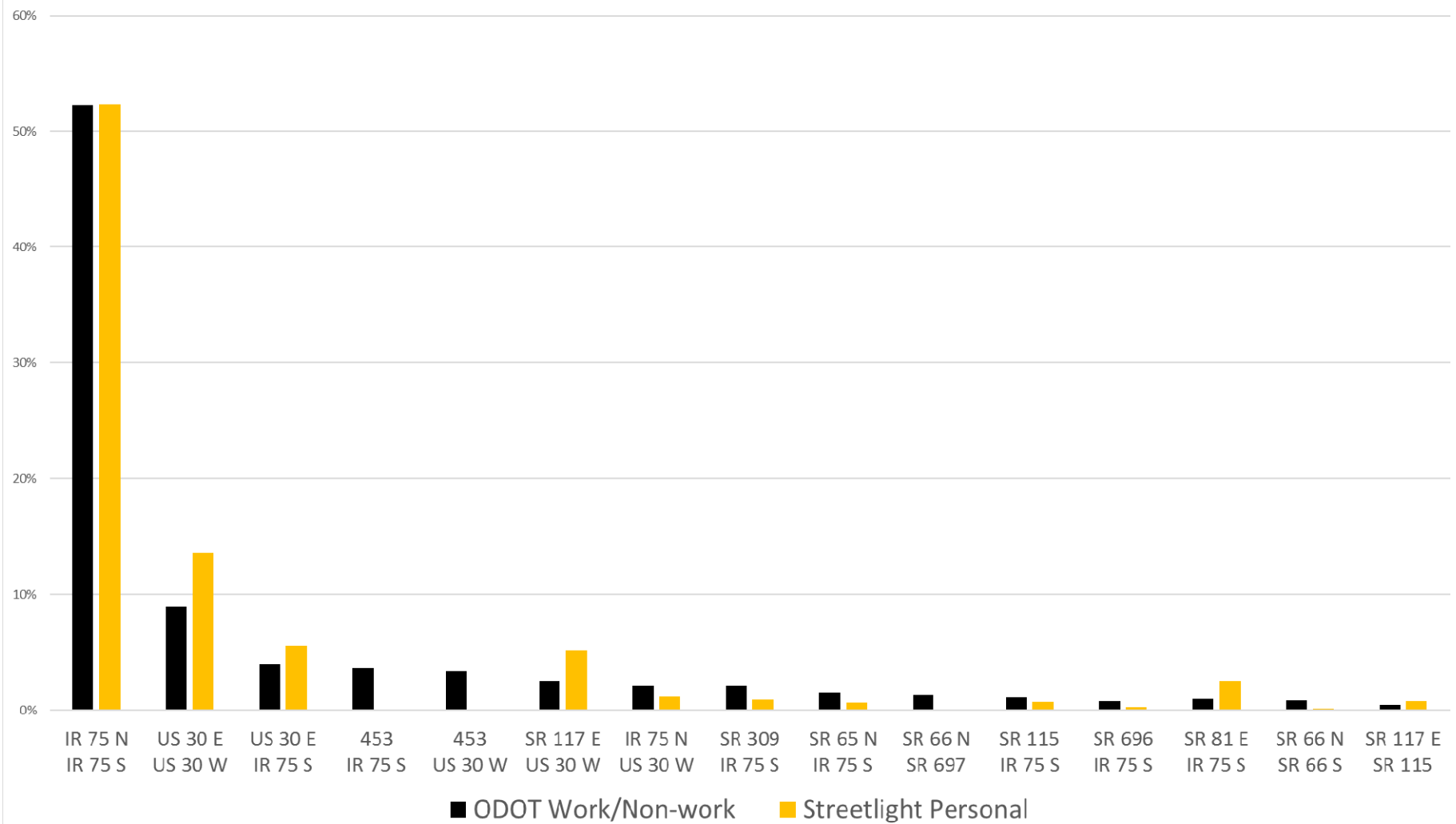
EE flow - all vehicles (15 Highest % flows)

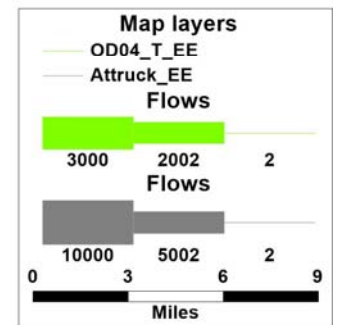
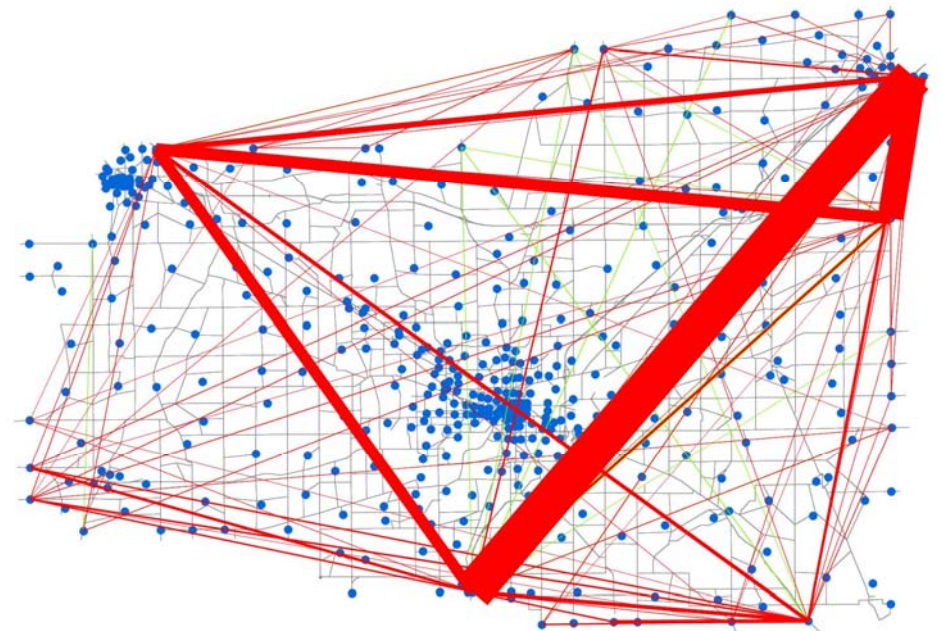
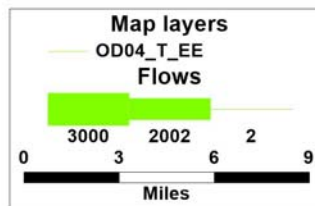
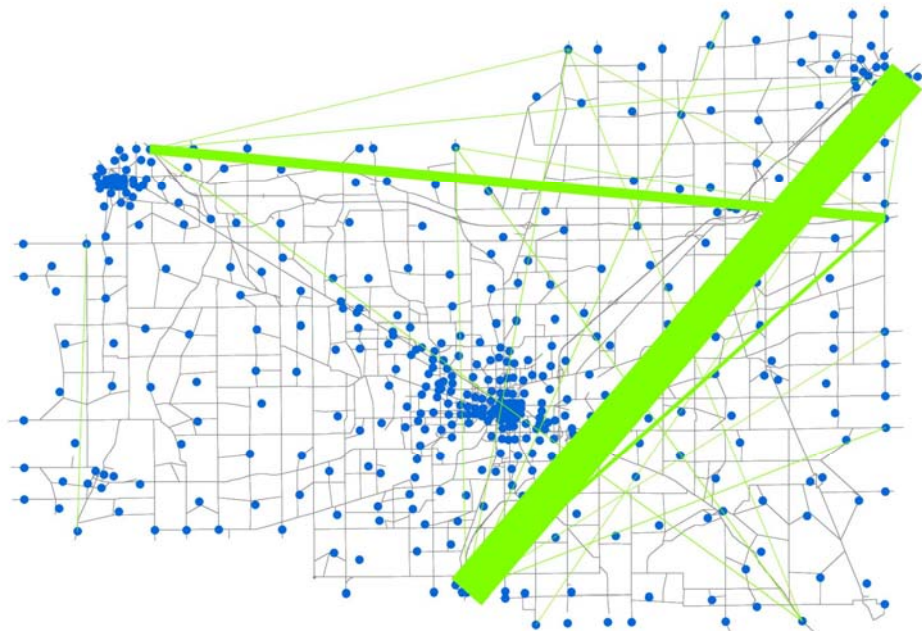


EE flow - Truck (15 Highest % flows)



EE flow - Personal (15 Highest % flows)







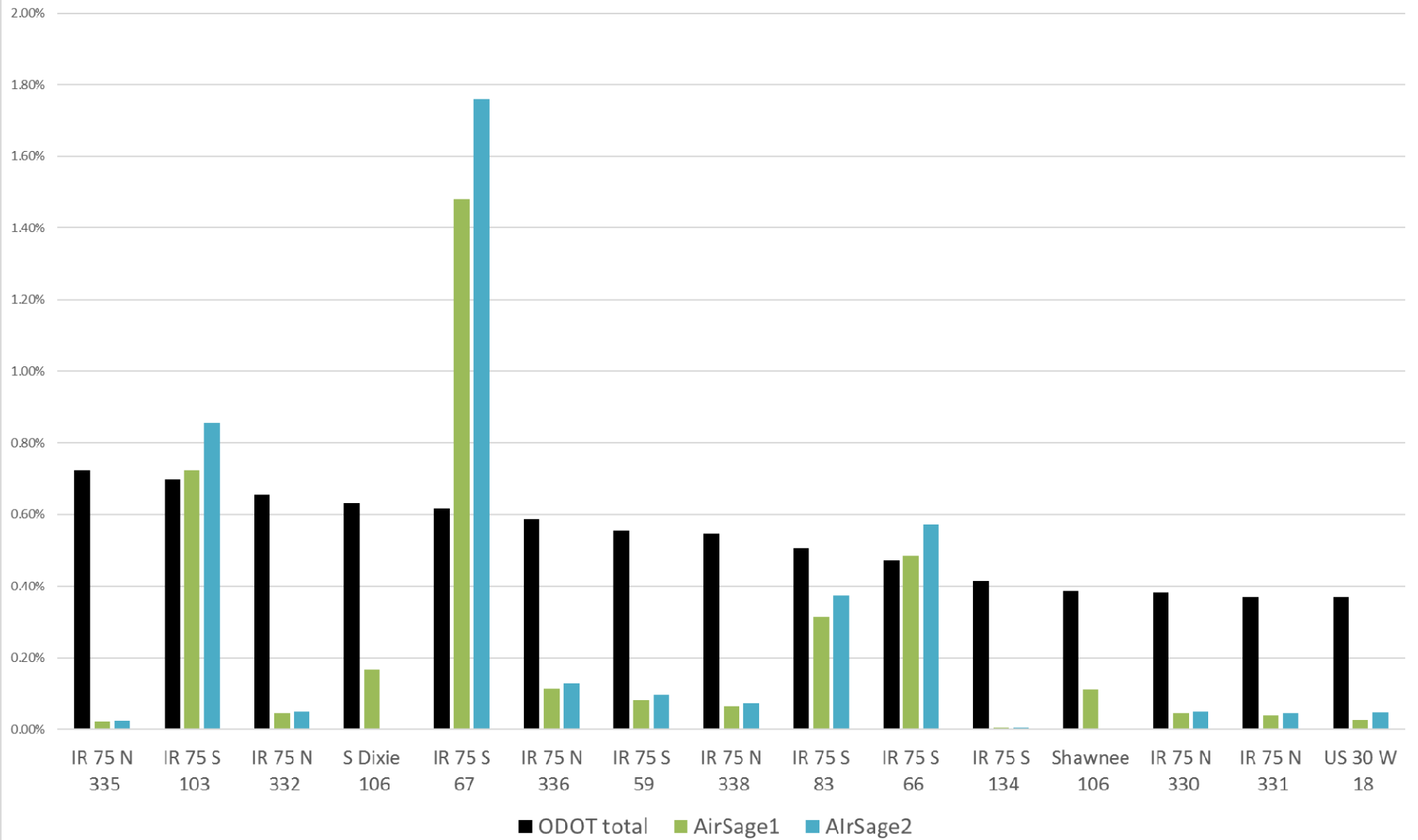
4. EI/IE flows

Table 2 Overall goodness of fit measures (AME, RSME and R square)

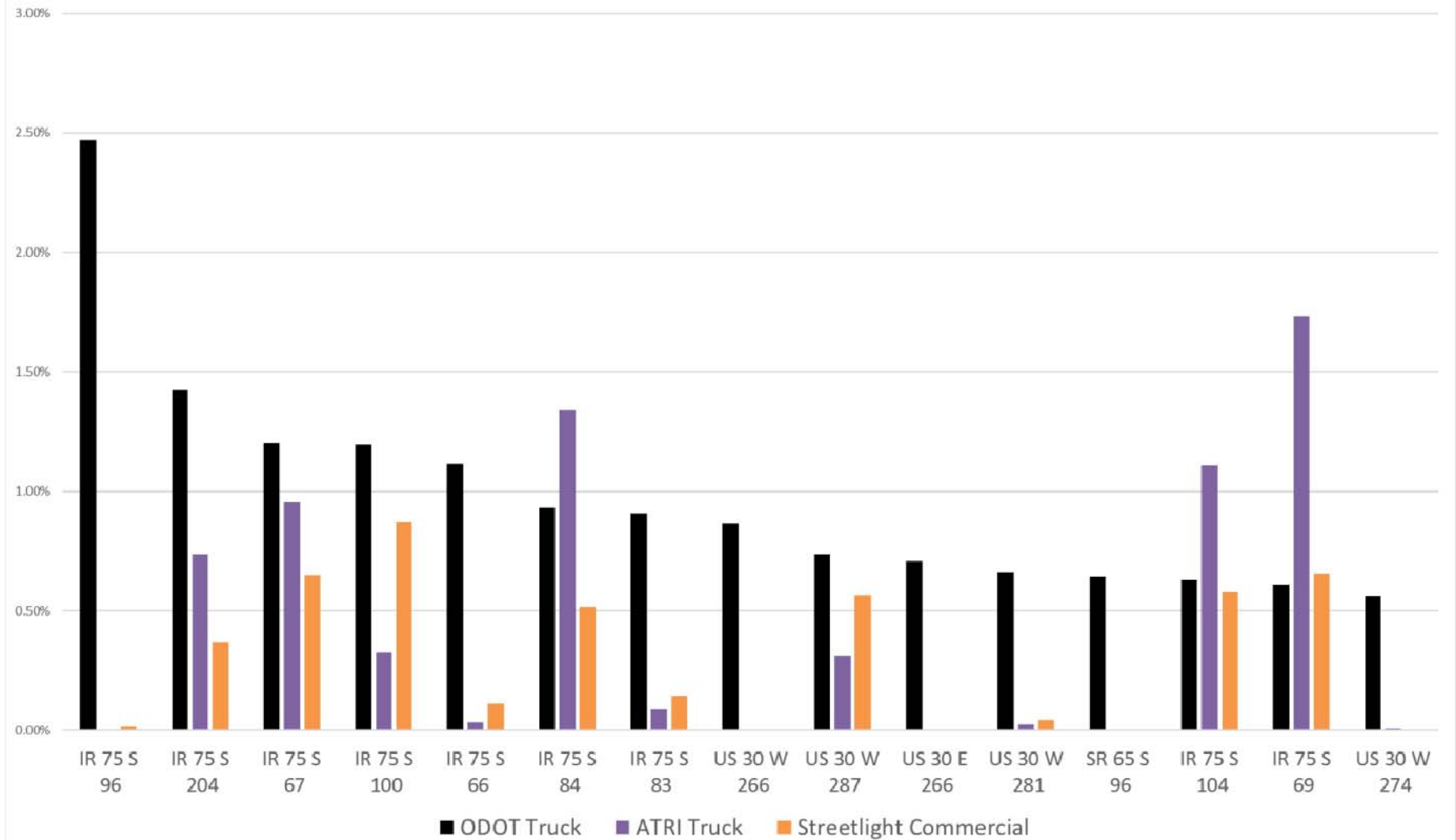
Ground-truth	ATD	Absolute mean error	Root mean square error	R Square
ODOT 2011 (total)	<u>AirSage</u> ADT weighted	5.0	21.2	0.12
ODOT 2011 (total)	<u>AirSage</u> Shortest path modeled	5.3	22.6	0.10
ODOT 2011 Truck	ATRI	6.7	123.4	0.02
ODOT 2011 Truck	Streetlight Commercial	47.9	1012.5	0.01
ODOT 2011 Non-work + work	Streetlight Personal	29.0	627.8	0.01

Both absolute and relative fit measures are poor, indicating that ATD fails to replicate EI/IE trip distribution of ODOT data

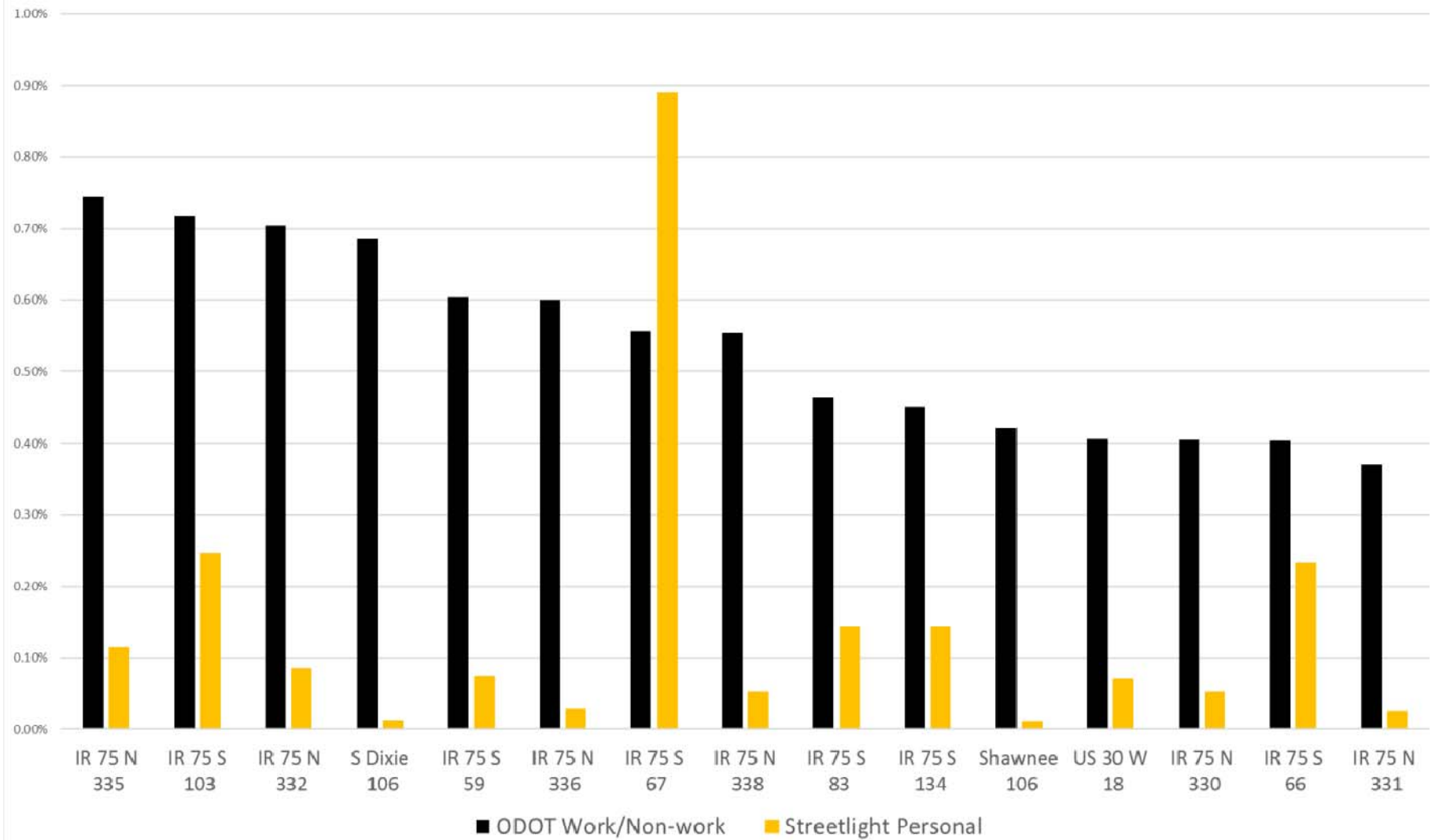
EI/IE flows - all vehicles (15 highest % flows)

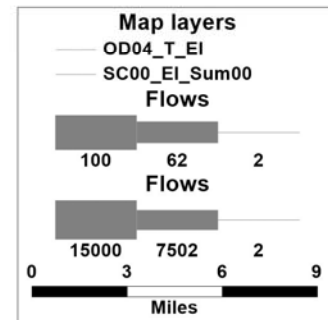
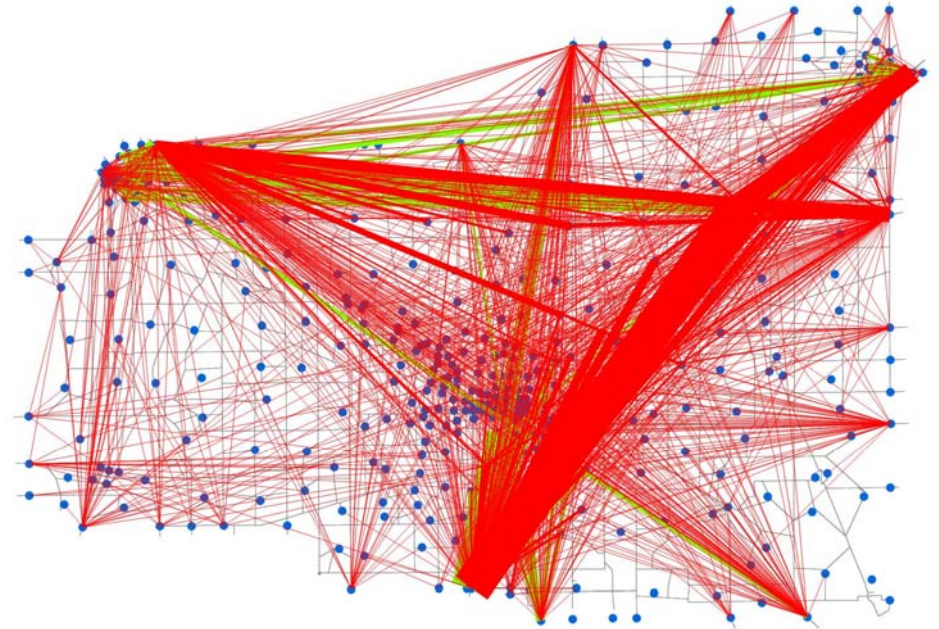
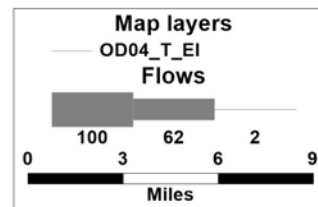
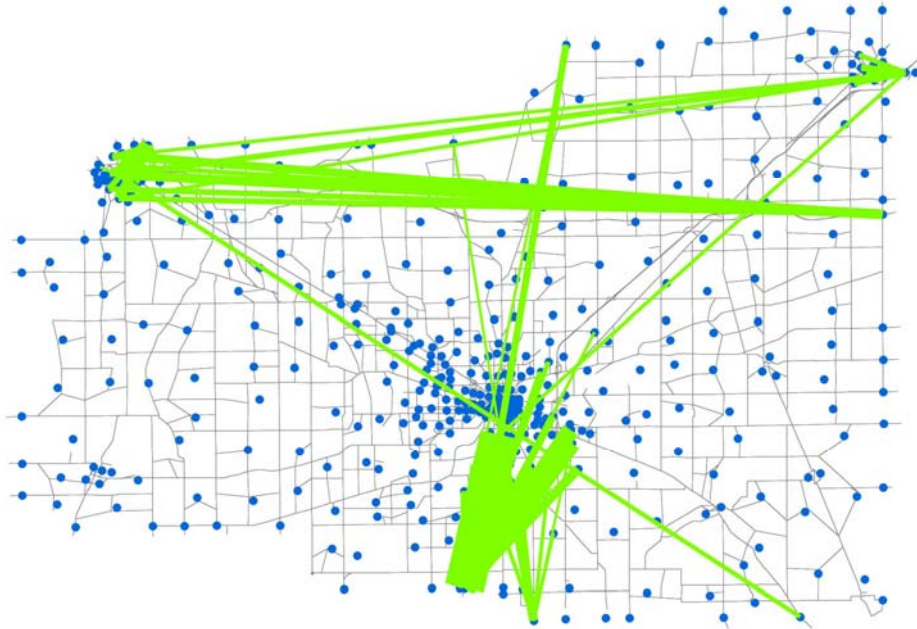


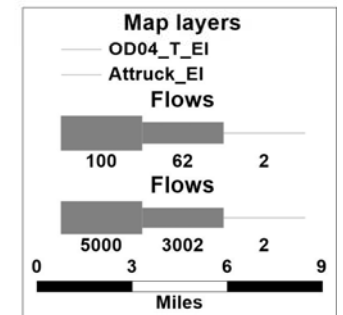
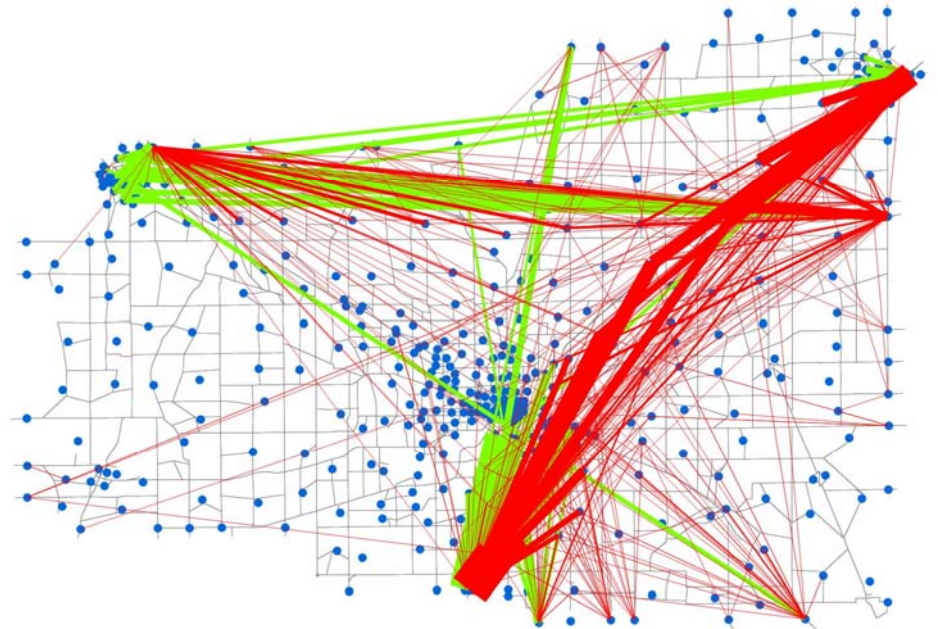
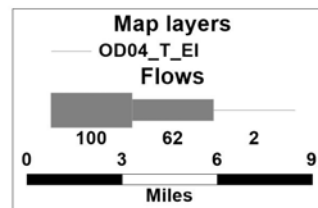
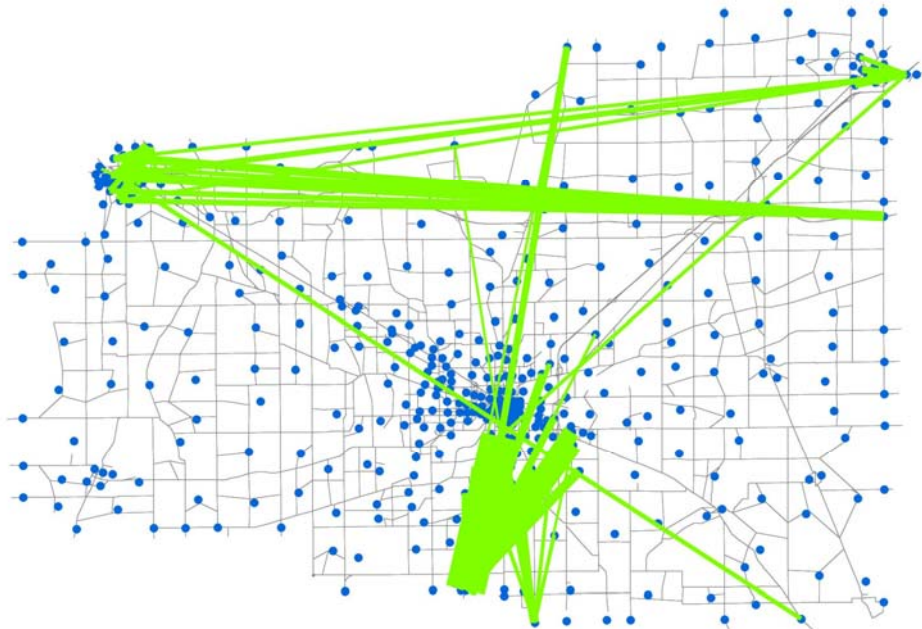
El/IE flows - Truck (15 highest % flows)



EI/IE flows - Personal (15 highest % flows)









4. EI/IE flows

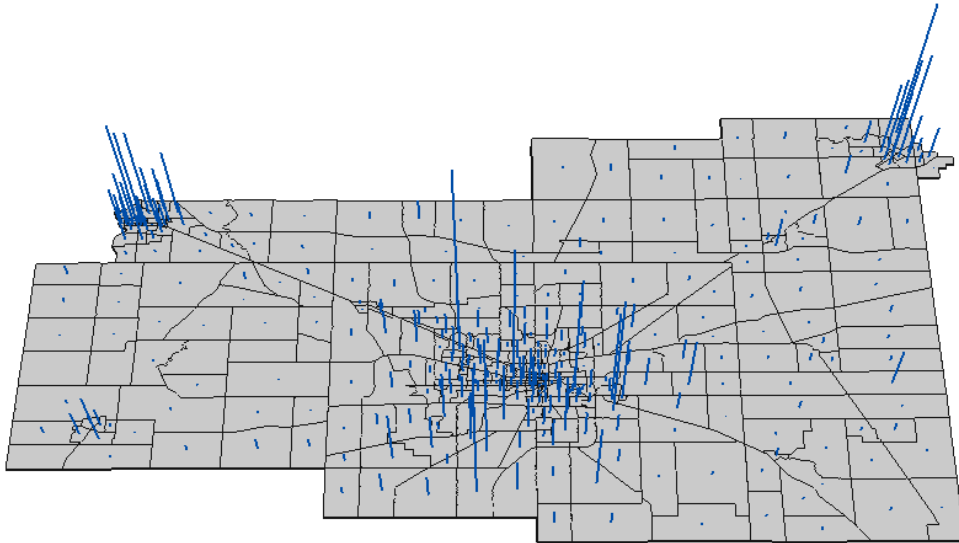
TAZ-based analysis

The following slides show the percentage of traffic originating from and destined to TAZs

General pattern

ODOT data: TAZs with a high percentage of traffic tend to be located in urban locations

ATD: TAZ with a high percentage of traffic volume are along major highways (especially in ATRI Truck, Streetlight commercial and personal data).

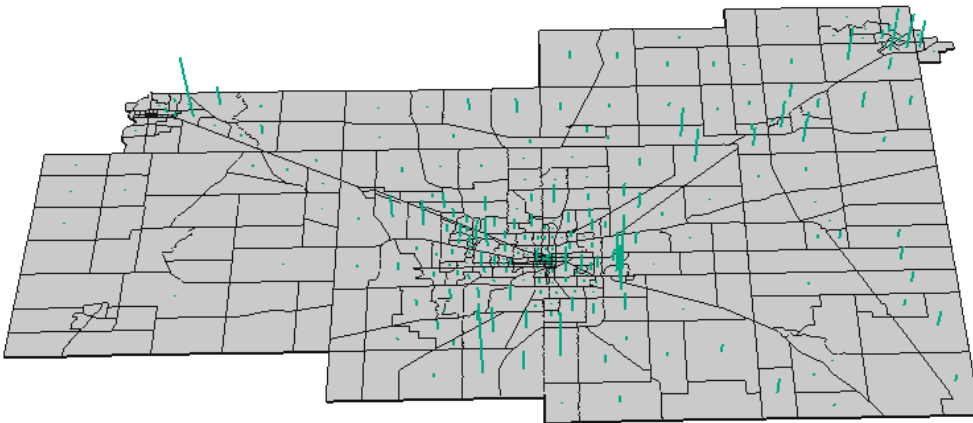


ODOT Total

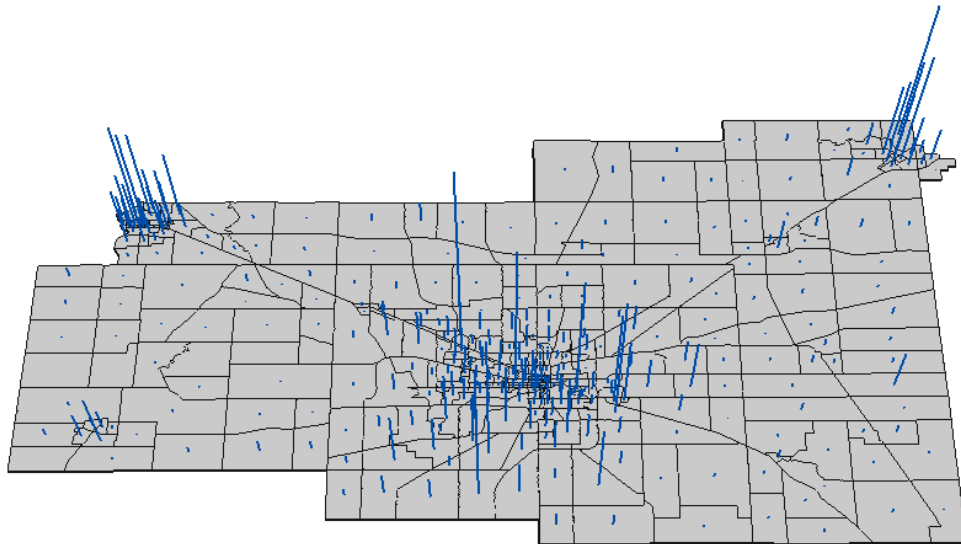
Absolute mean error: 5.0

Root mean square error: 21.2

R Square: 0.12



Airsage1 (weighted)

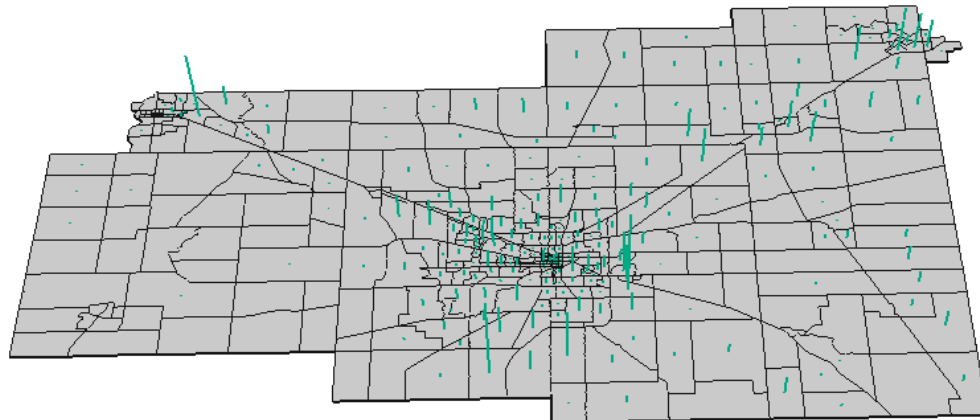


ODOT Total

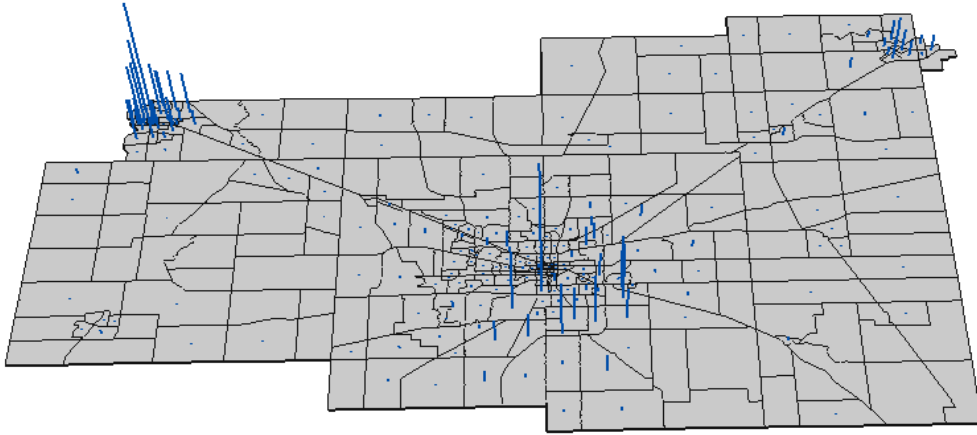
Absolute mean error: 5.3

Root mean square error: 22.6

R Square: 0.10



Airsage2 (shortest path)

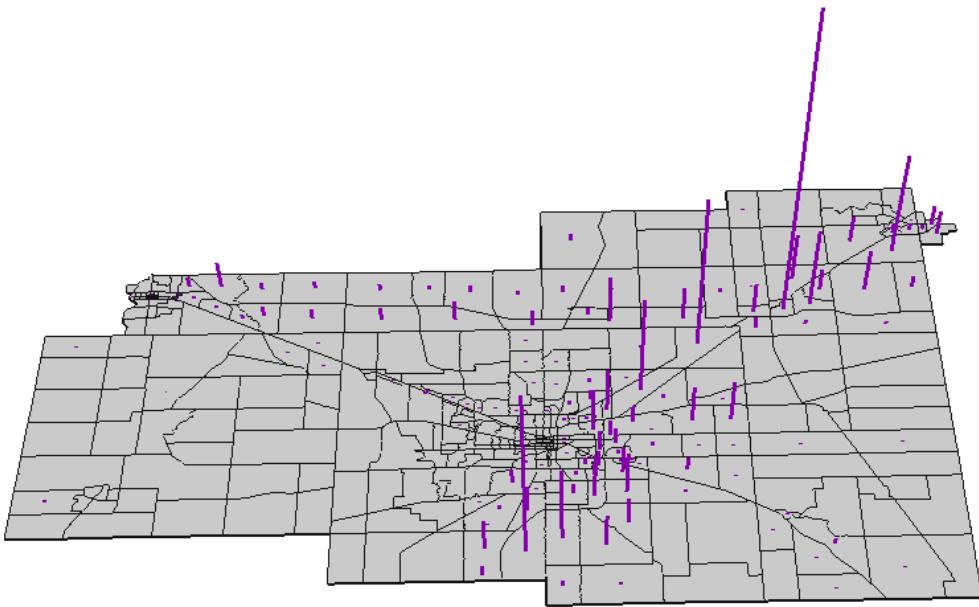


ODOT Truck

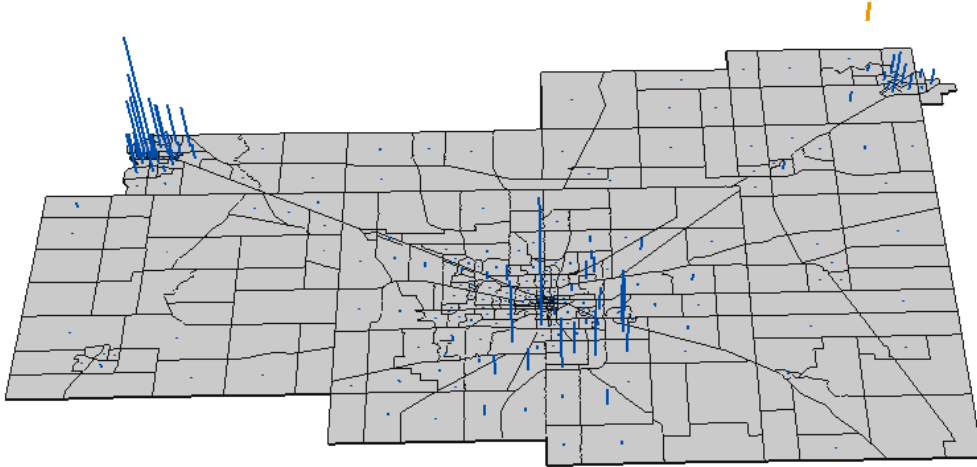
Absolute mean error: 6.7

Root mean square error: 123.4

R Square: 0.02



ATRI Truck

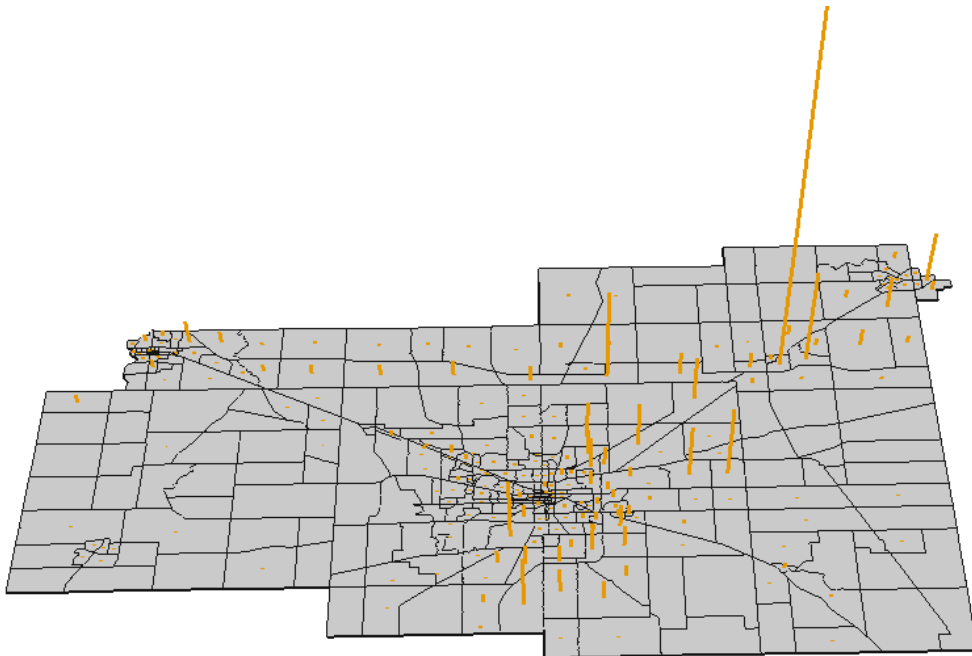


ODOT Truck

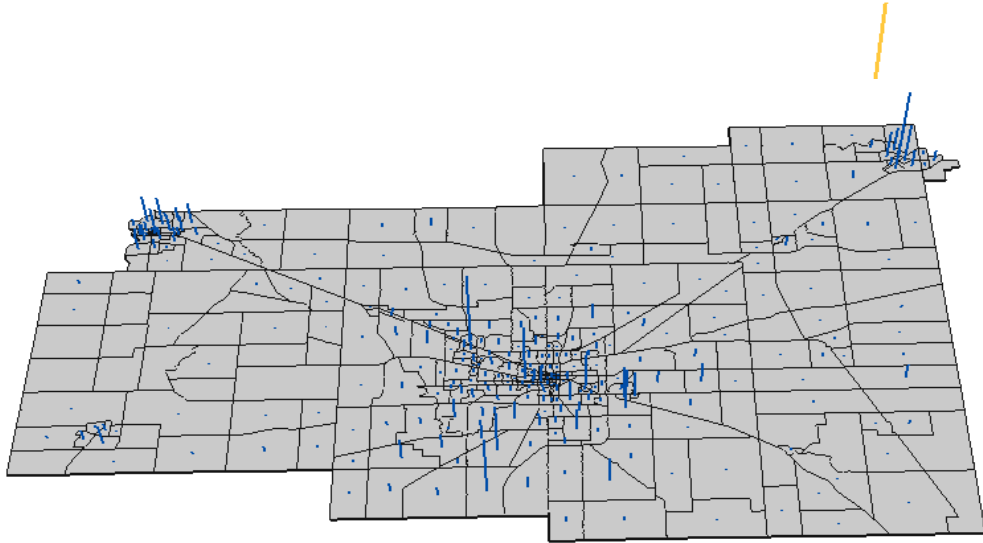
Absolute mean error: 47.9

Root mean square error: 1012.5

R Square: 0.01



Streetlight Commercial

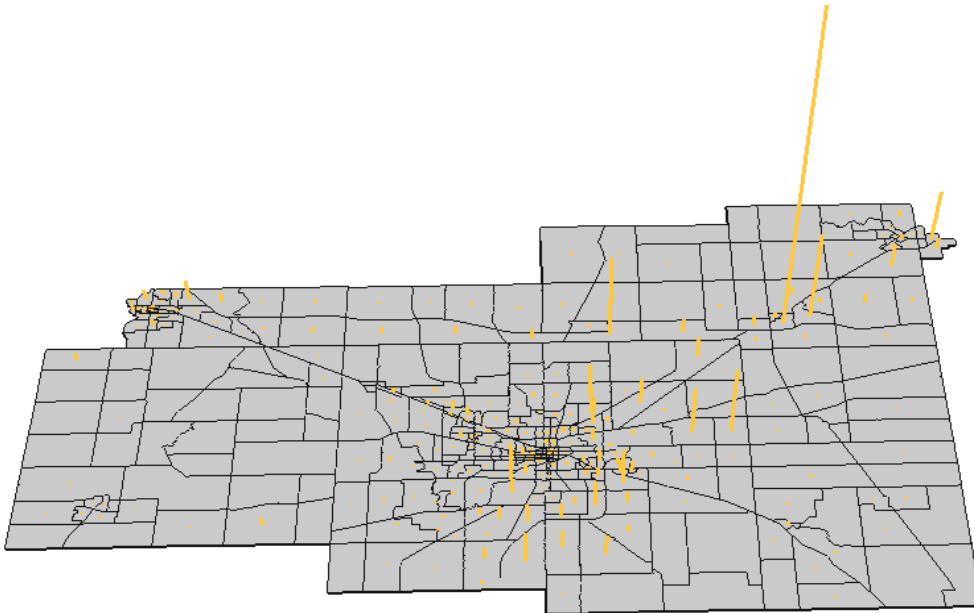


ODOT Work/Nonwork

Absolute mean error: 29.0

Root mean square error: 627.8

R Square: 0.01



Streetlight Personal



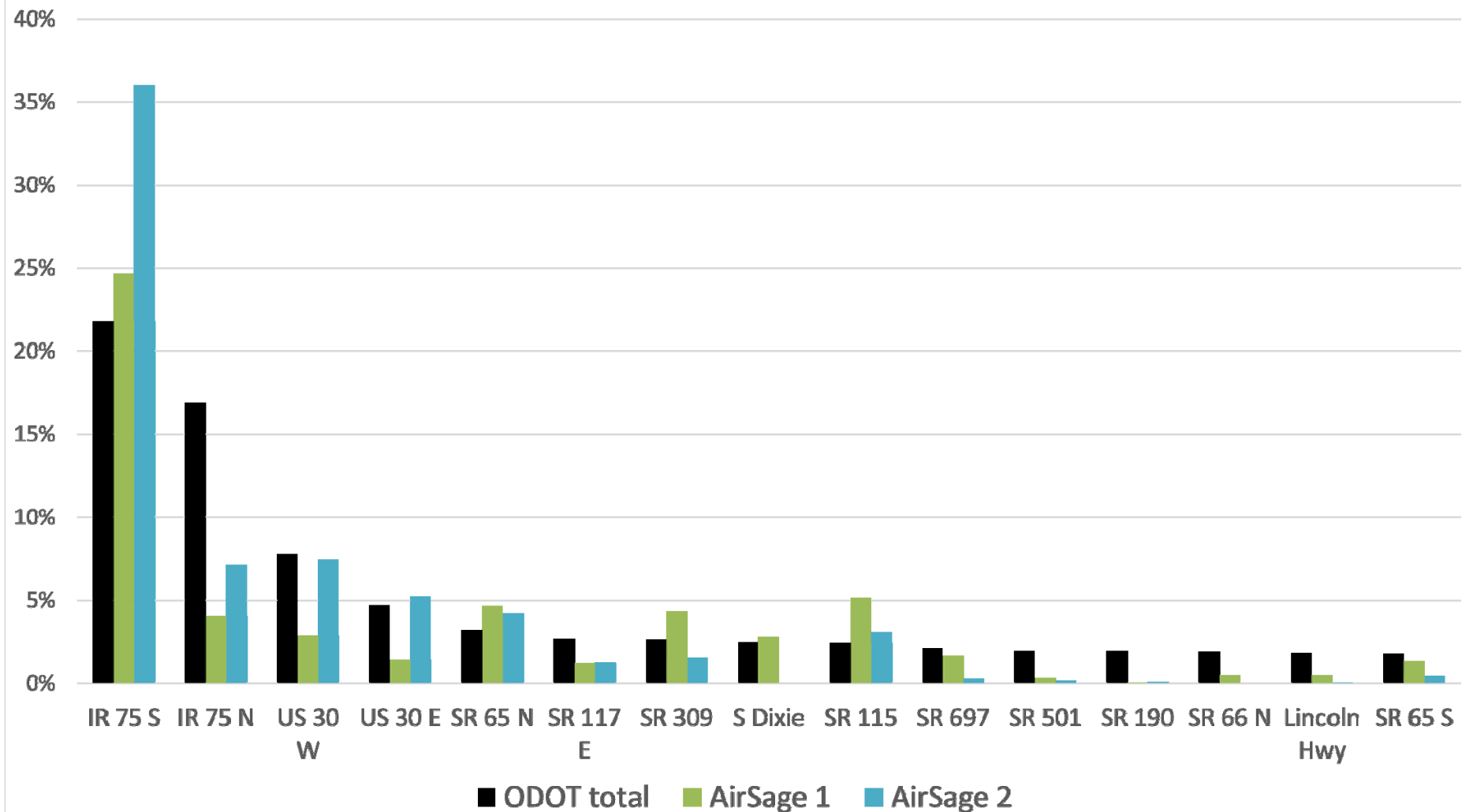
5. External stations

Table 2 Overall goodness of fit measures (AME, RSME and R square)

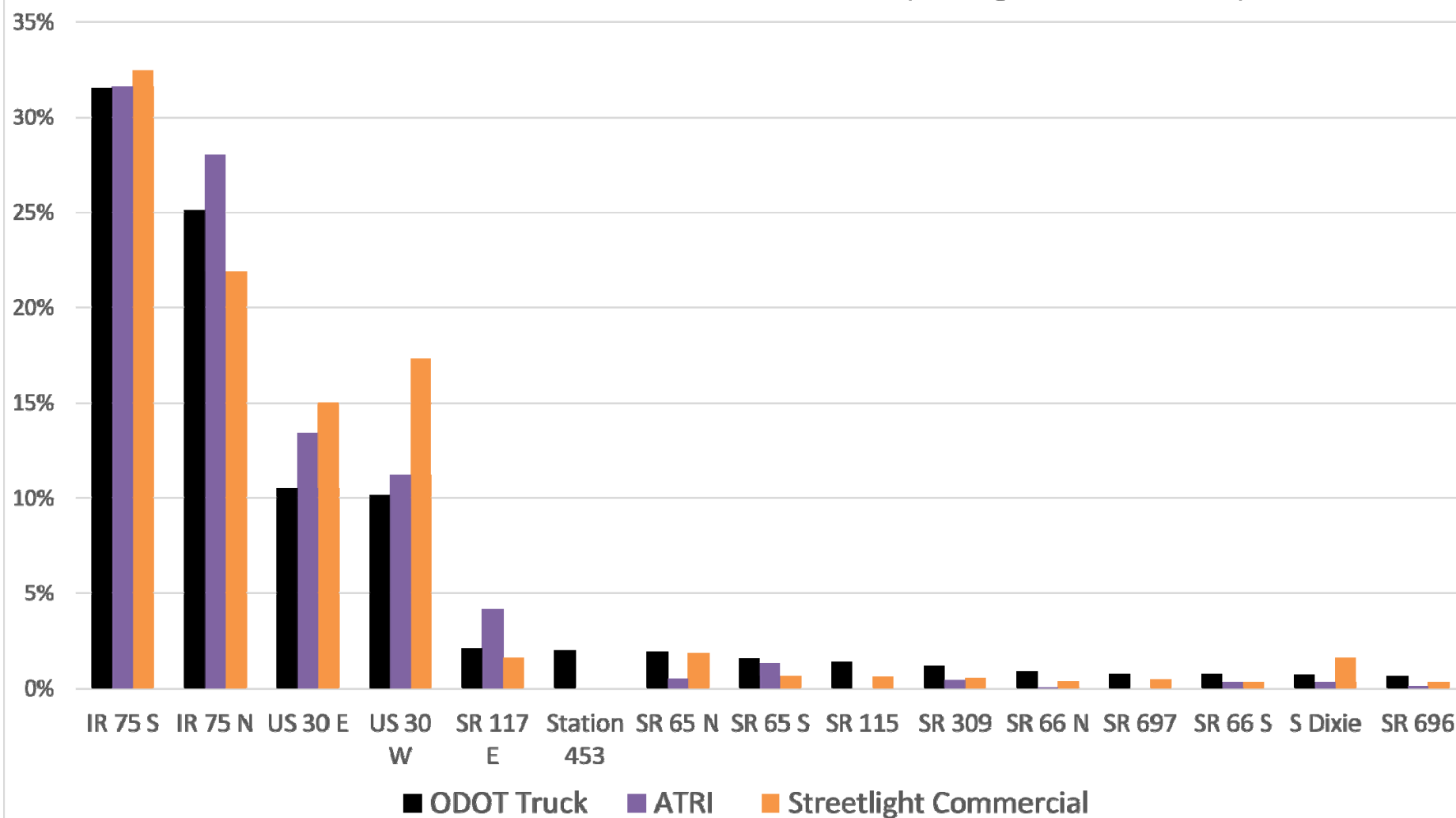
Ground-truth	ATD	Absolute mean error	Root mean square error	R Square
ODOT 2011 (total)	<u>AirSage</u> ADT weighted	2181.6	4657.5	0.57
ODOT 2011 (total)	<u>AirSage</u> Shortest path modeled	2301.4	4729.8	0.72
ODOT 2011 Truck	ATRI	4048.0	13482.8	0.99
ODOT 2011 Truck	Streetlight Commercial	64940.3	217405.6	0.95
ODOT 2011 Non-work + work	Streetlight Personal	7916.4	25760.2	0.94

Absolute error measures suggest poor fit (due to scale differences)
Relative measures suggest better fit

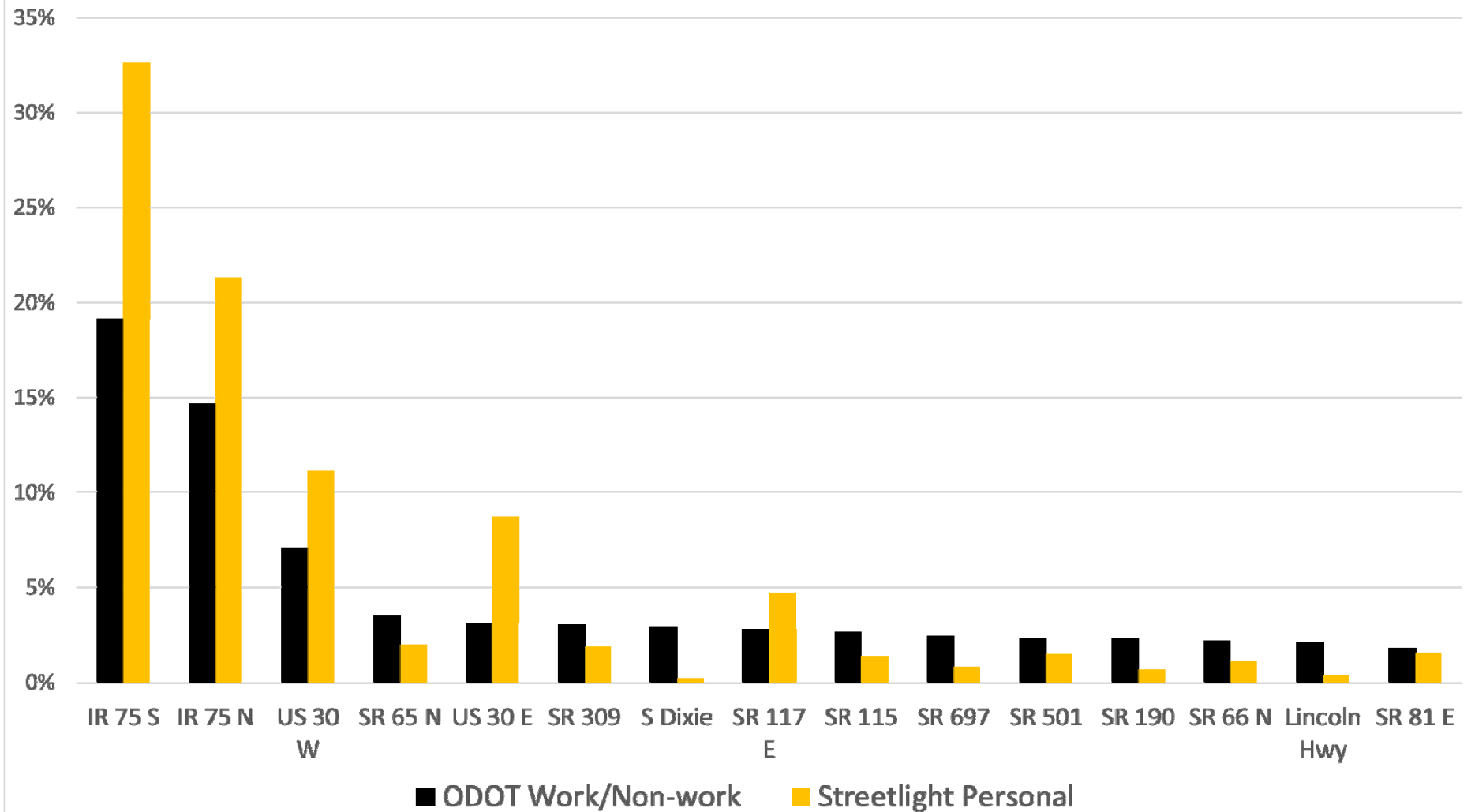
Traffic volume at external stations - All vehicles (15 Highest % stations)



Traffic volume at external stations - Truck (15 Highest % stations)



Traffic volume at external stations - Personal (15 Highest % stations)





Summary

	AirSage 1	AirSage2	ATRI	Streetlight Commercial	Streetlight Personal
1. Trip length	Fair	Good	Fair	Good	Good
2. Trip purpose					
3. EE flows	Poor	Good	Fair	Good	Good
4. IE flows	Poor	Poor	Poor	Poor	Poor
5. External stations	Fair	Fair	Good	Good	Good



Conclusions (preliminary)

1. ATD has more data with more general coverage
2. Need methods to reconcile data scale across different studies
3. Overall: ATRI and Streetlight data have better fit
4. AirSage: shortest path assignment improves fit
5. ATD fit to ODOT IE flows and TAZ totals is poor



Thank you! Questions?

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- Center for Urban and Regional Analysis: <https://cura.osu.edu/>